

## # Force

In physics, a **force** is a push or pull acting upon an object as a result of the object's interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

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### ## Key Characteristics

- **Vector Quantity** – Forces have both magnitude (strength) and direction.
- **SI Unit** – The standard metric unit for force is the **Newton (N)**.

One Newton is the force required to accelerate a 1 kg mass at a rate of  **$1 \text{ m s}^{-2}$** .

- **Formula** – Newton's Second Law of Motion:

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$$F = m a$$

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where  $(F)$  is force,  $(m)$  is mass, and  $(a)$  is acceleration.

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### ## Common Types of Forces

Forces are generally divided into two main categories:

#### ### 1. Contact Forces

Applied through physical contact between objects.

Examples:

- **Friction** – resistance to relative motion.
- **Tension** – pulling force transmitted through a string or cable.

- **Air resistance** – drag force acting on moving objects.
- **Normal force** – perpendicular reaction force from a surface.
- **Applied push/pull** – direct human or mechanical forces.

### ### 2. Non-Contact (Field) Forces

Act over a distance without physical contact.

Examples:

- **Gravitational force** – attraction between masses.
- **Electrostatic force** – attraction/repulsion between charged particles.
- **Magnetic force** – interaction between magnetic fields and currents.
- **Nuclear forces** – strong and weak interactions binding atomic nuclei.

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These notes capture the essential concepts of force, its mathematical description, and the primary categories of forces encountered in physics.